ATMOSYNC

Micro-Climate Arbitrage Analytics

4-Week Project Roadmap & Work Distribution

A practical 20-working-day plan for a four-member Data Analytics team

Duration	4 weeks / 20 working days
Team	4 members
Mid-review	End of Week 2 — Day 10
Final review	End of Week 4 — Day 20
Core stack	Python, Pandas, NumPy, Matplotlib/Seaborn, Streamlit or Power BI, Git/GitHub

Purpose

This plan divides AtmoSync into four weekly phases, assigns ownership to each member and schedules a mid-review after the first 10 working days. It is a practical proposed schedule aligned with the Data Analytics assignment scope.

Important scope note: The assignment describes data collection, cleaning, analysis, visualization, dashboards, insights and documentation. It does not prescribe a specific arbitrage formula or require machine learning. Any opportunity score in this plan is a team-defined analytical indicator, not a verified financial profit measure.

1. Four-member role allocation

Each member owns a workstream but should participate in peer review, integration, GitHub updates and both presentations.

Member	Primary responsibility	Work and outputs
1	Data collection & research	Understand data requirements; select locations/date range; retrieve Open-Meteo historical weather data; inspect W60 workbook; document sources, variables, units and limitations. Outputs: raw data, source log, data dictionary, collection notebook.
2	Data quality & preprocessing	Profile data; standardize timestamps and data types; inspect duplicates, missing values and outliers; document treatment; prepare analysis-ready tables. Outputs: quality report, cleaned dataset, preprocessing notebook.
3	EDA & analytical insights	Calculate descriptive statistics; analyze time trends; compare temperature/humidity; make spatial comparisons only where data supports them; define a transparent opportunity indicator. Outputs: EDA notebook, charts, findings and indicator definition.
4	Dashboard, integration & reporting	Build dashboard in Streamlit or Power BI; integrate outputs; coordinate report, PPT, README and demo. Outputs: dashboard, documentation, presentations and demo checklist.

Shared responsibilities

All four members attend a 10–15 minute daily check-in, update the task tracker, commit their work to GitHub, review a teammate's output, record blockers and participate in reviews and the final demo.

Recommended stack

Core: Python, Jupyter Notebook/VS Code, Pandas, NumPy, Matplotlib/Seaborn, Git and GitHub. **Dashboard:** choose either Streamlit or Power BI. **Data:** Open-Meteo Historical Weather API and the supplied W60 benchmark workbook, subject to suitability.

2. Week 1 — Requirements, sourcing and dataset understanding

Goal: define the analytical question, establish the repository, obtain source data and understand its structure.

Day	Task / expected activity	Owner(s)
Day 1	Kick-off; interpret assignment; define problem statement, objectives, scope and roles.	All members; Member 4 creates GitHub repository/task board.
Day 2	Select study locations and date range; list variables, units and source limitations.	Member 1 leads; Members 2–3 review.
Day 3	Retrieve historical weather data; inspect W60 workbook sheets, columns, timestamps and shipment IDs.	Member 1 leads; Member 2 supports.
Day 4	Create data dictionary; profile rows, columns, missing values, duplicates and time coverage.	Member 2 leads; Member 1 verifies metadata.
Day 5	Save raw-data snapshot; document gaps/risks; provide weekly progress update.	All members; Member 4 consolidates.

Deliverables: GitHub repository, scope/problem statement, raw data, source log, data dictionary and initial data-quality profile.

Checkpoint: Can the team explain what each dataset represents, its coverage, units, missingness and limitations?

3. Week 2 — Cleaning, preparation and mid-review

Goal: produce a reproducible cleaned dataset and show credible initial analysis by Day 10.

Day	Task / expected activity	Owner(s)
Day 6	Standardize timestamps/time zones, column names and data types; preserve raw files unchanged.	Member 2 leads; Member 1 validates.
Day 7	Assess duplicates, missing values and outliers; document treatment and avoid unjustified imputation.	Member 2 leads; Member 3 reviews impact.
Day 8	Create analysis-ready tables; calculate summary statistics and initial time-series plots.	Members 2–3; Member 4 drafts dashboard layout.
Day 9	Analyze temperature/humidity distributions and trends; prepare valid comparisons.	Member 3 leads; Member 1 checks location metadata; Member 4 prepares review slides.
Day 10	Present mid-review: source data, cleaning workflow, initial charts, issues and next steps.	All members present their contributions.

Deliverables: cleaned dataset, preprocessing notebook, cleaning log, initial EDA charts, mid-review PPT and feedback/action list.

MID-REVIEW — Day 10: demonstrate working files and code, not only slides. Explain completed work, preliminary observations, data limitations, blockers and the remaining 10-day plan.

4. Week 3 — Deeper analysis and dashboard build

Goal: develop defensible comparisons and a usable first dashboard.

Day	Task / expected activity	Owner(s)
Day 11	Apply mid-review feedback; revise scope or cleaning where required; record decisions.	All members; Member 2 updates preprocessing.
Day 12	Analyze hourly/daily patterns, temperature range, humidity variation and relationships.	Member 3 leads; Member 2 supplies validated tables.
Day 13	Compare locations/time periods only where data is aligned; define/test a transparent opportunity indicator.	Member 3 leads; Member 1 validates location context.
Day 14	Build dashboard pages for KPIs, trends, comparisons and filters; connect cleaned data.	Member 4 leads; Members 2–3 validate figures.
Day 15	Integrate analysis and dashboard; conduct walkthrough; record defects and missing items.	All members; Member 4 coordinates.

Deliverables: EDA findings, valid comparisons, documented indicator (if supported), first integrated dashboard and defect list.

Checkpoint: Every chart and KPI should trace to a documented calculation and data field. Label estimates and limitations clearly.

5. Week 4 — Validation, documentation and final demonstration

Goal: verify results, finish project artifacts and deliver a repeatable demonstration.

Day	Task / expected activity	Owner(s)
Day 16	Validate calculations, joins, filters, missing-data handling and dashboard totals against source tables.	Members 2–3 lead; Members 1 and 4 cross-check.
Day 17	Improve dashboard labels, units, tooltips and layout; resolve priority defects.	Member 4 leads; all members test.
Day 18	Complete report: objective, sources, methodology, cleaning, analysis, findings and limitations.	All members contribute; Member 4 consolidates.
Day 19	Finalize README, requirements, code organization, PPT and demo script; rehearse clean run.	All members; Member 4 coordinates.
Day 20	Final review and live demonstration; explain contributions, results, limitations and future work.	All members present.

Deliverables: validated dashboard, final data, source code/notebooks, README, report, PPT and final demo.

FINAL REVIEW — Day 20: demonstrate the workflow from data input to insights/dashboard, explain limitations and show the GitHub repository.

6. Review presentation checklists

Mid-review — end of Week 2 (Day 10)

• Project title, problem statement, objectives and scope

• Team roles and work completed by each member

• Dataset sources, date range, location coverage and variable descriptions

• Raw dataset structure and data-quality findings

• Cleaning/preprocessing steps and reasons

• Initial EDA charts and cautious observations

• Challenges, data limitations and mitigation plan

• Plan for Weeks 3 and 4

Final review — end of Week 4 (Day 20)

• End-to-end data collection and preprocessing workflow

• Analysis methodology, KPI definitions and reproducible findings

• Working dashboard with filters and clearly labeled units

• Interpretation of findings and actionable recommendations

• Limitations, assumptions and future improvements

• GitHub repository, README and run instructions

• Final report, PPT and live demonstration

• Clear explanation of each member's contribution

7. Suggested GitHub folder structure

```
AtmoSync/
■■■ data/
■ ■■■ raw/ # Original source files; do not overwrite
■ ■■■ processed/ # Validated analysis-ready data
■■■ notebooks/
■ ■■■ 01_data_collection.ipynb
■ ■■■ 02_data_quality_cleaning.ipynb
■ ■■■ 03_exploratory_analysis.ipynb
■■■ src/ # Reusable Python functions
■■■ dashboard/
■ ■■■ app.py # If using Streamlit
■■■ reports/ # Final report and figures
■■■ presentation/ # Mid-review and final PPT
■■■ README.md
■■■ requirements.txt
■■■ .gitignore
```

Repository hygiene: keep raw files unchanged; do not commit API keys, credentials or sensitive data. Use approved private storage for large or restricted datasets.

8. Technical safeguards

• **Weather data resolution:** Open-Meteo historical products may be model/reanalysis-based, not direct measurements at every site. State the source and resolution accurately.

• **Fair spatial comparison:** compare aligned timestamps, consistent units and documented coordinates. If there is only one location, report temporal patterns rather than spatial arbitrage.

• **W60 missingness:** profile missing values by sensor position and shipment. Do not blindly fill columns with extensive gaps.

• **Target leakage:** inspect engineered or future-looking columns before using predictors. Exclude information that would not be available at decision time.

• **Meaning of arbitrage:** the assignment does not define a financial arbitrage formula. A team-defined score can summarize observed differences, but it is not a profit estimate without operational/cost evidence.

• **Machine learning:** do not force a prediction model. Validated descriptive analytics and a working dashboard are appropriate core deliverables.

9. Weekly status tracker

Week	Milestone	Evidence	Status
1	Scope and data ready	Repo, raw data, source log, data dictionary, quality profile	Not started / In progress / Done
2	Clean data + mid-review	Cleaning notebook, processed data, initial charts, review PPT	Not started / In progress / Done
3	Analysis + dashboard alpha	EDA findings, indicator definition, first dashboard	Not started / In progress / Done
4	Validated final delivery	Testing, report, README, PPT, final demo	Not started / In progress / Done

Daily coordination rhythm

Daily: 10–15 minute stand-up. **Twice weekly:** peer review code and findings. **Every Friday:** demonstrate weekly outputs and update the tracker. **Before reviews:** test from a clean checkout/environment.

This is a suggested execution plan aligned with the assignment scope and 20-working-day duration. Confirm exact review dates, submission formats and any additional mentor instructions with Infotact Solutions.